An Empirical Analysis of Correlation Between Ownership of Cultural Venues and Public Art Installations

By Fernando Bonilla, Matthew Lim, Alastair Nicholson, Mollie Sohn, and Natasha Taylor

This paper examines the relationship between cultural venue ownership and the density of public art installations, focusing on how private and government-owned venues shape urban cultural landscapes. Using neighborhood-level data from Vancouver, we developed and tested regression models incorporating socio-economic controls, including average household income and population density. Contrary to our null hypothesis that private ownership would negatively correlate with public art density due to profit-driven motives, our findings reveal a positive and statistically significant relationship between private cultural venues and public art installations. This effect is particularly pronounced in high-density neighborhoods, where population density amplifies the influence of private venues on public art proliferation. Robustness checks, including interaction and nonlinear control models, validated these results and highlighted the complex interplay between density, socio-economic factors, and venue ownership. These findings challenge traditional assumptions about public art being predominantly driven by government initiatives and underscore the critical role of private institutions in cultivating cultural vibrancy in urban spaces. The results provide valuable insights for policymakers balancing public and private investments in cultural infrastructure to enhance community identity and inclusivity.

The density of public art installations reflects urban quality of life, enhancing cityscapes and contributing to social and economic well-being. Public art fosters tourism, cultural engagement, and inclusivity, making it a key part of cultural infrastructure. This paper investigates how the ownership of cultural venues influences public art density in surrounding areas.

Prior research highlights a fundamental distinction between government-owned and privately owned cultural venues. Publicly owned museums and galleries are typically guided by

objectives of inclusivity and community access, with funding often directed toward maximizing public benefit (Kawashima, 1997). Conversely, privately owned venues may prioritize profitability and cater to more niche audiences. These contrasting missions suggest potential differences in their contributions to the local cultural landscape, particularly regarding the proliferation of public art installations. Based on these distinctions, we hypothesized a negative association between private cultural venues and public art density, under the assumption that private ownership would prioritize profit-driven activities over investments in public cultural assets.

To address this research question, we developed and tested two regression model approaches to explore the relationship between venue ownership and public art density, using varying controls for neighborhood wealth. The first model incorporated average household income, revealing a negative relationship between wealthier neighborhoods and public art density, suggesting a potential preference for privatized land use over public art. The second model introduced income per capita and highlighted an interaction between private ownership and population density as a critical factor influencing public art distribution.

Our findings challenge the null hypothesis. Across both models, private cultural venues demonstrated a positive and statistically significant association with public art density, even after controlling for economic and demographic factors. This suggests that, contrary to the assumption that private ownership limits public art proliferation, the presence of private venues may actively contribute to an increase in public art installations. While these results leave room for varying interpretations—such as public art installations attracting private cultural spaces or private ownership cultivating more public art—this paper focuses on establishing the empirical relationship between the two.

I. Significance

Economic prosperity serves as a fundamental pillar for the well-being and advancement of any society. Analyzing the intricate relationship between economic growth and cultural investment allows policymakers to evaluate the extent to which subsidies for public cultural venues contribute to public art accessibility, tourism, and economic activity. Moreover, exploring the spill-over effects of public gallery funding, as opposed to reliance on privately-funded counterparts, provides critical insights into how cultural investments can produce both direct and indirect socio-economic outcomes. This understanding informs decisions on resource allocation, ensuring that cultural

policies contribute to both economic and social development. Additionally, the positive relationship depicted in our results between private cultural venues and public art installations prompts policy makers to consider alleviating policy structure around additional private cultural venue formulation to promote urban cultural development.

II. Data Description

We assembled a neighbourhood-level dataset to explore how ownership of cultural venues influences the density of public art installations in surrounding areas. Our dataset covers the City of Vancouver and draws from multiple sources, including records of public art installations and information about cultural spaces through surveys in their given neighbourhood.

A. Cultural Venues

The City of Vancouver provided a neighborhood-level dataset as part of the city's cultural spaces map pilot project, which is updated annually in June through stakeholder surveys within the cultural community. This dataset includes detailed records on cultural spaces such as museums and galleries across various neighborhoods, with each entry specifying the venue's location (local area) and ownership type.

The distinction between public and private ownership is crucial, as ownership type may influence the allocation of resources to cultural initiatives and the level of engagement with the local community. Neighborhood location data enables an analysis of the spatial distribution of these venues, helping determine whether the presence of publicly or privately owned cultural spaces correlates with higher densities of public art installations in surrounding areas. Public ownership often reflects government investment or subsidies, potentially encouraging cultural activities as part of a government-driven initiative. Conversely, private ownership may indicate market-driven funding, which might discourage public art installations if perceived as reducing profitability. To simplify the analysis, venues not explicitly listed as either privately or government-owned were classified under public ownership, encompassing those identified as "other."

This study focuses on neighborhood-level analysis to capture localized effects, operating on the assumption that cultural spaces exert the most significant influence on public art installations within their immediate vicinity. Additionally, the dataset highlights neighborhoods lacking cultural venues, providing a baseline for comparison and a clearer picture of accessibility to cultural resources.

B. Public Art Installations

The City of Vancouver also supplied a public art dataset, updated weekly, detailing installations across various neighborhoods. Each record includes the precise location of the installation, allowing the aggregation of total installations by neighborhood. Public art installations are critical to this study as they reflect a neighborhood's cultural vibrancy and engagement with public spaces.

This dataset forms the foundation for measuring the density of public art within each neighborhood. It is assumed that all recorded installations are publicly accessible and relevant to the local community. This assumption is essential for understanding how cultural engagement materializes in a spatial context.

C. Census Data

The City of Vancouver's neighborhood-level census dataset provides key variables such as average total income and population density. These variables serve as controls in our analysis, enabling a comprehensive exploration of how ownership of cultural venues impacts the density of public art installations in a neighborhood.

Average total income serves as a proxy for economic resources and community investment, both of which often influence public art initiatives. Population density captures the concentration of residents, potentially shaping community engagement and the visibility of public installations. By incorporating these socio-economic and demographic controls, the study isolates the impact of ownership type on public art density while accounting for broader contextual factors.

D. Variable Interactions

It is important to note that income and population density, while serving as controls, may also directly influence the ownership of cultural venues, which is our dependent variable. For instance, wealthier neighborhoods may attract more private cultural investments, while denser neighborhoods might motivate public funding for cultural spaces to meet the demand for community engagement. These interdependencies suggest that income and population density could share overlapping explanatory power, potentially introducing multicollinearity into the model.

III. Summary Statistics

Table: Summary Statistics for Merged Data (Public Art, Cultural Spaces, and Income)

Neighbourhood	ownership_group	total_public_art	avg_sq_ft	total_spaces	avg_household_income
Downtown	Government	138	47157.787	155.0	76700
Downtown	Privately Owned	74	20027.667	138.0	76700
Grandview-Woodland	Government	20	10437.037	27.0	85900
Grandview-Woodland	Privately Owned	1	3834.800	50.0	85900
Kensington-Cedar Cottage	Government	16	88554.000	10.0	95100
Kensington-Cedar Cottage	Privately Owned	6	10000.000	5.0	95100
Kerrisdale	Privately Owned	1	96172.852	33.1	91600
Kitsilano	Government	21	47848.171	41.0	96100
Kitsilano	Privately Owned	3	4000.000	3.0	96100
Marpole	Government	4	4968.636	11.0	98500
Marpole	Privately Owned	12	3917.857	14.0	98500
Mount Pleasant	Government	32	9385.531	32.0	86600
Mount Pleasant	Privately Owned	25	4321.021	96.0	86600
Renfrew-Collingwood	Government	7	5318.333	21.0	91000
Renfrew-Collingwood	Privately Owned	1	5522.500	8.0	91000
RileyPark	Government	8	96172.852	33.1	114100
RileyPark	Privately Owned	1	96172.852	33.1	114100
South Cambie	Government	4	96172.852	33.1	124900
Stanley Park	Government	28	36456.600	20.0	72400
Sunset	Government	5	2264.000	4.0	94000
Sunset	Privately Owned	1	96172.852	33.1	94000
West End	Government	23	28829.800	15.0	74000
West End	Privately Owned	19	2500.000	4.0	74000
West Point Grey	Government	3	65182.250	20.0	118050

TABLE 1—MODEL 1 SUMMARY STATISTICS

Summary Statistics

Statistic	Mean	Median	St. Dev.	Min	Max	N
Public_Art_Count	26.773	11.5	52.305	0	249	22
Private	43.136	8	70.964	0	240	22
AI	54,739.500	50,156.5	21,142.880	31,534	118,668	22
PD	28,778.860	29,250	15,188.880	7,970	62,030	22

TABLE 2— MODEL 2 SUMMARY STATISTICS

IV. Regression Model

A. Model 1

(1) $$Y_i = \beta^0 + \beta^1 D_i + \beta^2 PD + \beta^3 AI + \varepsilon_i$$

Simple Regression Model.—For the first regression analysis, we examined two linear models to measure the relationship between the number of public art installations and the density of cultural spaces, specifically focusing on the impact of privately owned versus government-owned cultural spaces. The first model employed a simple linear regression to establish a baseline understanding of this relationship.

(2) $$Y_i = \beta^0 + \beta^1 * D_i * PD + \beta^2 * AI + \beta^3 * D_i PD * AI + \varepsilon_i$$

Complex Regression Model.—The second, more complex model incorporated interaction terms to capture the interplay between explanatory variables and their combined effects on the dependent variable. Our objective was to determine whether ownership type influences the presence of public art installations in a neighborhood or interacts significantly with other explanatory variables in the model.

Key Variables

- D_i : Dummy variable where $D_i = 1$ represents privately owned spaces and $D_i = 0$ represents government-owned spaces.
- PD: Density of cultural spaces, calculated as the total number of spaces divided by the average square footage.
- AI: Average household income, also aggregated by neighborhood, to control for economic factors at the community level.

B. Model 2

(3) $$Y_i = \beta_0 + \beta_1 Private_i + \varepsilon_i$$

Simple Regression Model.—We aimed to examine the direct impact of private venue density on the number of public art installations in a neighborhood. Our initial hypothesis was that an increase in private venues might reduce public art installations, driven by the profit-oriented nature of businesses. Public art, being freely accessible, could attract people outdoors for leisure rather than inside cultural venues, diminishing the appeal of profit-based spaces. To explore this, we started with a simple model to identify any correlation. However, comparing the effects of private venues across neighborhoods required accounting for other factors that might influence this relationship.

(4) $$Y_i = \beta_0 + \beta_1 Private_i + \beta_2 AI_i + \beta_3 PD_i + \varepsilon_i$$

Revised Model.—The revised model accounts for two key neighborhood factors: Average Total Income per capita (AI) and Population Density (PD). We believe these are important controls. AI may attract private businesses to areas with disposable income, so we expect a positive

relationship with private cultural venues. As for public art (Y_i), its relationship with AI is uncertain.

Higher AI might encourage the government and artists to foster a positive neighborhood spirit, benefiting mental health and GDP, suggesting a positive relationship with Y_i. However, higher AI could also mean more privatized land, limiting public art installations.

Regarding PD, higher population density may prompt the government to support public spaces for social interaction, which could influence the amount of public art. These controls are essential for our regression, where we examine the effect of private cultural venues on public art in a neighborhood.

(5) $$Y_i = \beta_0 + \beta_1 Private_i + \beta_2 AI_i + \beta_3 PD_i + \beta_4 AI^2_i + \beta_5 PD^2_i + \varepsilon_i$$

Nonlinear Control Model.—This is part of the robustness test for the revised regression model. We consider the revised model final, as it shows the precise relationship we expected. The robustness test examines whether any nonlinear effects with the controls are missing. We added β_4 and β_5 to explore these potential relationships. The test checks if the relationship between private cultural venues and public art (Y_i) remains the same as before. The controls help account for any potential omitted variable bias (OVB) affecting this relationship.

(6) $$Y_i = \beta_0 + \beta_1 Private_i \times PD_i + \beta_2 AI_i + \varepsilon_i$$

Interaction Model.—This model is part of the robustness test for the revised regression. We wanted to confirm if the relationship between private cultural venues and public art (Y_i) is truly causal, not just a correlation. While the original revised model showed an interesting result, we felt it lacked deeper analysis. The interaction term Private:PD looks at how these two influence the relationship each has with Y_i. Although we considered adding an interaction term for Private and AI, we excluded it because AI showed minimal significance in both the revised and first

robustness test models, compared to Private and PD. The β_1 in this model is designed to look at exactly that.

(7) $$Y_i = \beta_0 + \beta_1 Private_i \times PD_i + \beta_2 PD^2_i + \varepsilon_i$$

Private-PD Model.—This model is the culmination of all previous attempts to make a better model. The two robustness tests showcase four findings: 1. AI is not important to the relationship. 2. PD works better as a nonlinear control. 3. The interaction term between Private and PDis highly significant. 4. Private alone doesn't explain Y_i considering the interaction term. We found that the key variables are Private:PD and PD^2. The β_1 here will determine whether Private:PD has a positive or negative effect on Y_i. The β_2 determines the positive or negative relationship PD² has with the public art count, highlighting its parabolic relationship with Y_i at low and high PD levels.. As the β_1 number alone isn't interpretable alone, we will just be seeing the signs it proposes with this model.

Key Variables

- Y_i: $Number\ of\ Public\ arts\ in\ neighbourhood\ i$
- $Private_i$: $Number\ of\ Private\ cultural\ venues\ in\ neighbourhood\ i$
- AI_i: $Average\ Total\ Income\ per\ capita\ in\ neighbourhood\ i.$
- PD_i: $Population\ density\ in\ neighborhood\ i.$

V. Results

A. Model 1

```
Call:
lm(formula = total_public_art ~ density + avg_household_income +
    ownership_dummy, data = merged_data_cleaned_filtered)

Residuals:
    Min      1Q  Median      3Q     Max
-23.135 -14.050  -5.199   5.274  94.357

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)          1.228e+02  4.040e+01   3.039  0.00648 **
density              6.420e+02  1.218e+03   0.527  0.60395
avg_household_income -1.059e-03  4.153e-04  -2.551  0.01903 *
ownership_dummy      -1.602e+01  1.158e+01  -1.383  0.18188
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 26.82 on 20 degrees of freedom
Multiple R-squared:  0.3069,    Adjusted R-squared:  0.203
F-statistic: 2.952 on 3 and 20 DF,  p-value: 0.05739
```

FIGURE 3.1 SIMPLE REGRESSION RESULTS

Notes: The only significant variables in our model are the Average Household Income and the Ownership Dummy

** Significant at the 1 percent level.

* Significant at the 5 percent level.

```
Call:
lm(formula = total_public_art ~ density * avg_household_income *
    ownership_dummy, data = merged_data_cleaned_filtered)

Residuals:
    Min      1Q  Median      3Q     Max
-22.553  -6.086  -2.075   1.781  26.784

Coefficients:
                                            Estimate Std. Error t value
(Intercept)                               -3.027e+01  3.735e+01  -0.810
density                                    2.141e+05  3.452e+04   6.204
avg_household_income                       5.098e-04  3.744e-04   1.362
ownership_dummy                            4.193e+01  7.071e+01   0.593
density:avg_household_income              -2.362e+00  3.894e-01  -6.066
density:ownership_dummy                   -1.611e+05  4.074e+04  -3.953
avg_household_income:ownership_dummy      -5.371e-04  7.413e-04  -0.724
density:avg_household_income:ownership_dummy  1.756e+00  4.623e-01   3.798
                                          Pr(>|t|)
(Intercept)                                0.42962
density                                    1.26e-05 ***
avg_household_income                       0.19217
ownership_dummy                            0.56143
density:avg_household_income               1.63e-05 ***
density:ownership_dummy                    0.00114 **
avg_household_income:ownership_dummy       0.47923
density:avg_household_income:ownership_dummy  0.00158 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 15.36 on 16 degrees of freedom
Multiple R-squared:  0.8182,    Adjusted R-squared:  0.7386
F-statistic: 10.29 on 7 and 16 DF,  p-value: 6.831e-05
```

FIGURE 3.2— COMPLEX REGRESSION RESULTS

B. Model 2

```
Robust Standard Errors (Simple Model):

t test of coefficients:

            Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.3230     5.1549  0.4506  0.65709
Private       0.5668     0.2385  2.3765  0.02758 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

FIGURE 4.1—SIMPLE REGRESSION MODEL POST RSE

```
Breusch-Pagan Test Results (Simple Model):

        studentized Breusch-Pagan test

data:  simple_model
BP = 14.598, df = 1, p-value = 0.000133
```

FIGURE 4.2—BP TEST ON SIMPLE MODEL

```
Robust Standard Errors (Revised Model):

t test of coefficients:

               Estimate  Std. Error t value Pr(>|t|)
(Intercept) -8.3006e+01  3.2960e+01 -2.5184 0.021470 *
Private      4.9236e-01  1.3972e-01  3.5239 0.002425 **
AI           8.0614e-04  2.9568e-04  2.7264 0.013852 *
PD           1.5432e-03  5.6078e-04  2.7519 0.013118 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

FIGURE 4.3—REVISED REGRESSION MODEL POST RSE

```
Variance Inflation Factors (Revised Model):
 Private       AI       PD
1.102457 1.212809 1.320739
```

FIGURE 4.4—REVISED REGRESSION MODEL VIF

```
Breusch-Pagan Test Results (Revised Model):

        studentized Breusch-Pagan test

data:  revised_model
BP = 14.293, df = 3, p-value = 0.002532
```

FIGURE 4.5—BP TEST ON REVISED MODEL

```
Call:
lm(formula = Public_Art_Count ~ Private + AI + PD + I(AI^2) +
    I(PD^2), data = final_data)

Residuals:
    Min      1Q  Median      3Q     Max
-43.634  -9.528   0.950  13.272  33.440

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.111e+01  4.155e+01   0.267  0.79259
Private      4.095e-01  7.099e-02   5.769 2.88e-05 ***
AI           6.924e-04  1.014e-03   0.683  0.50464
PD          -4.239e-03  1.521e-03  -2.788  0.01316 *
I(AI^2)     -2.809e-09  7.359e-09  -0.382  0.70770
I(PD^2)      8.747e-08  2.240e-08   3.904  0.00126 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 20.74 on 16 degrees of freedom
Multiple R-squared:  0.8802,    Adjusted R-squared:  0.8427
F-statistic:  23.5 on 5 and 16 DF,  p-value: 7.599e-07
```

FIGURE 4.6—ROBUSTNESS TEST ON NONLINEAR CONTROL MODEL

```
Call:
lm(formula = Public_Art_Count ~ Private * PD + AI, data = final_data)

Residuals:
    Min      1Q  Median      3Q     Max
-45.439  -5.579  -0.587   4.436  35.010

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -6.430e-01  2.452e+01  -0.026  0.97938
Private     -1.497e-01  1.481e-01  -1.011  0.32618
PD           9.916e-05  4.311e-04   0.230  0.82082
AI           7.618e-05  2.603e-04   0.293  0.77335
Private:PD   1.727e-05  3.646e-06   4.738  0.00019 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 18.46 on 17 degrees of freedom
Multiple R-squared:  0.8991,    Adjusted R-squared:  0.8754
F-statistic: 37.89 on 4 and 17 DF,  p-value: 2.939e-08
```

FIGURE 4.7—ROBUSTNESS TEST ON INTERACTION MODEL

```
Robust Standard Errors:

t test of coefficients:

               Estimate  Std. Error t value  Pr(>|t|)
(Intercept)  4.1112e+01  1.1305e+01  3.6368 0.0020394 **
Private     -2.1500e-02  1.1356e-01 -0.1893 0.8520842
PD          -3.1852e-03  9.9966e-04 -3.1863 0.0054048 **
I(PD^2)      5.6176e-08  1.9016e-08  2.9542 0.0088804 **
Private:PD   1.2454e-05  3.1188e-06  3.9931 0.0009413 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

FIGURE 4.8—PRIVATE-PD MODEL POST RSE

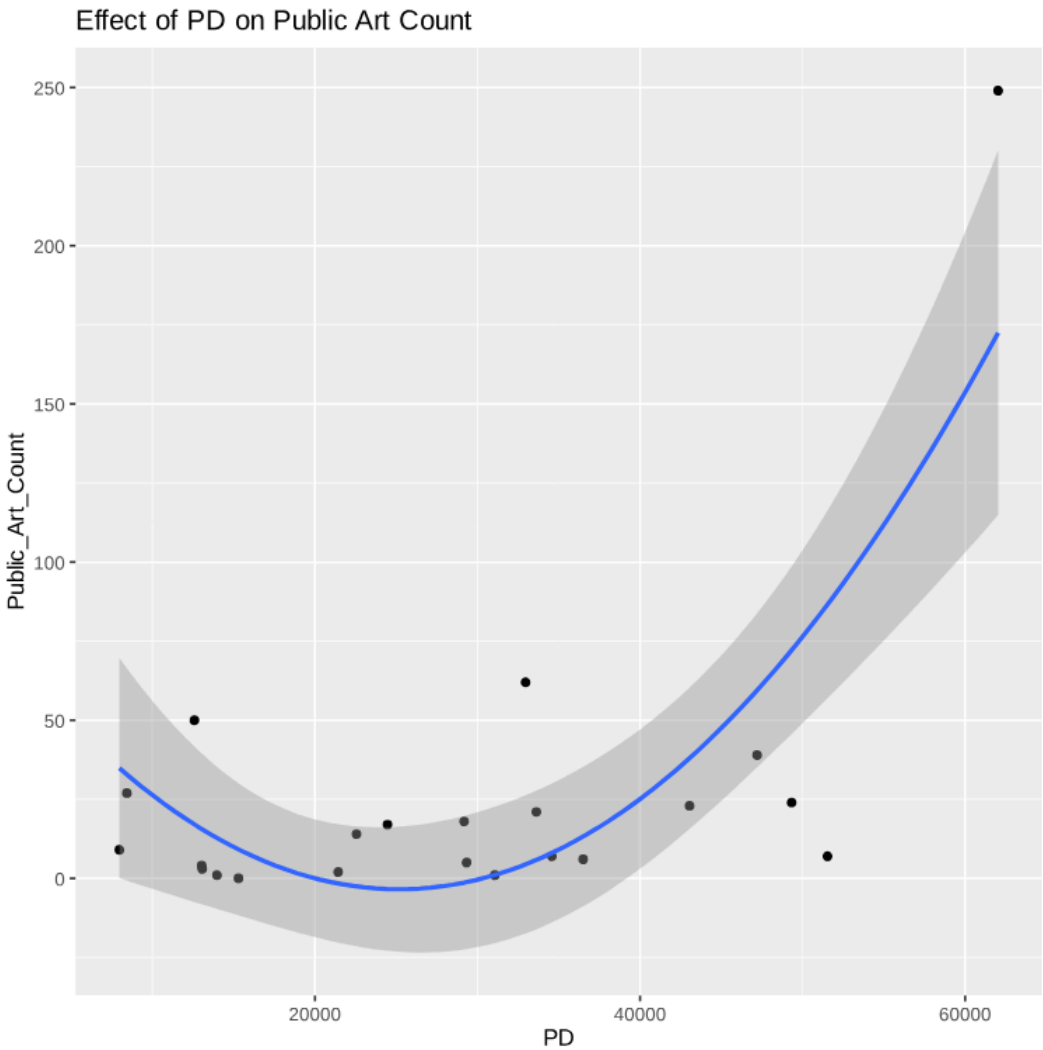

FIGURE 4.9—PD VS PUBLIC ART COUNT

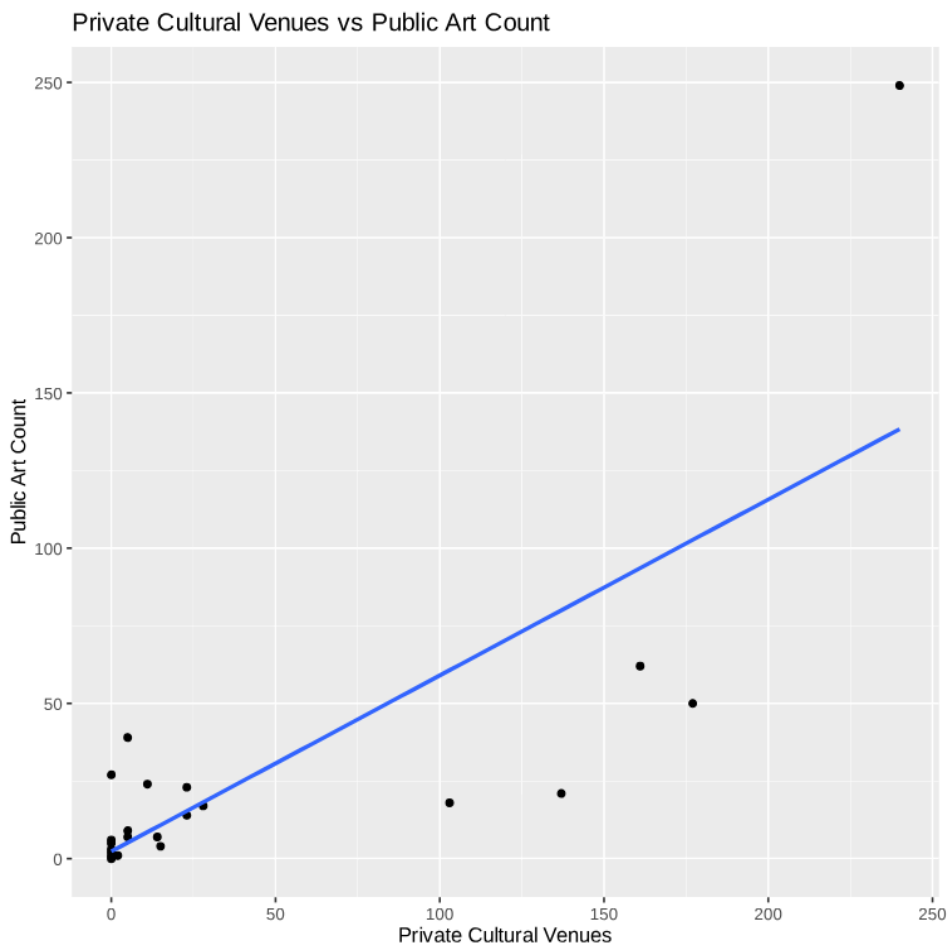

FIGURE 4.10—PRIVATE VS PUBLIC ART COUNT

```
High-Density Sub-sample Analysis:
Call:
lm(formula = Public_Art_Count ~ Private * PD + I(PD^2), data = high_density)

Residuals:
    Min      1Q  Median      3Q     Max 
-20.923  -3.885  -1.972   4.277  21.448 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)  
(Intercept) -1.137e+02  2.085e+02  -0.545   0.6054  
Private     -6.831e-01  3.841e-01  -1.778   0.1256  
PD           5.725e-03  1.072e-02   0.534   0.6125  
I(PD^2)     -6.383e-08  1.334e-07  -0.478   0.6493  
Private:PD   2.797e-05  9.940e-06   2.814   0.0306 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 16.01 on 6 degrees of freedom
Multiple R-squared:  0.9699,	Adjusted R-squared:  0.9499 
F-statistic: 48.38 on 4 and 6 DF,  p-value: 0.0001063
```

FIGURE 4.11—SUB SAMPLE ANALYSIS OF PRIVATE-PD MODEL

Notes: HIGH PD

```
Low-Density Sub-sample Analysis:
Call:
lm(formula = Public_Art_Count ~ Private * PD + I(PD^2), data = low_density)

Residuals:
     Min       1Q   Median       3Q      Max 
-10.0954  -1.6681  -0.1921   1.2799  11.7776 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)  
(Intercept)  6.658e+01  2.369e+01   2.810   0.0308 *
Private      6.402e-01  1.882e-01   3.403   0.0144 *
PD          -8.244e-03  3.319e-03  -2.484   0.0476 *
I(PD^2)      2.552e-07  1.070e-07   2.385   0.0544 .
Private:PD  -2.999e-05  1.323e-05  -2.266   0.0640 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 7.028 on 6 degrees of freedom
Multiple R-squared:  0.8676,	Adjusted R-squared:  0.7793 
F-statistic: 9.826 on 4 and 6 DF,  p-value: 0.008369
```

FIGURE 4.12—SUB SAMPLE ANALYSIS OF PRIVATE-PD MODEL

Notes: LOW PD

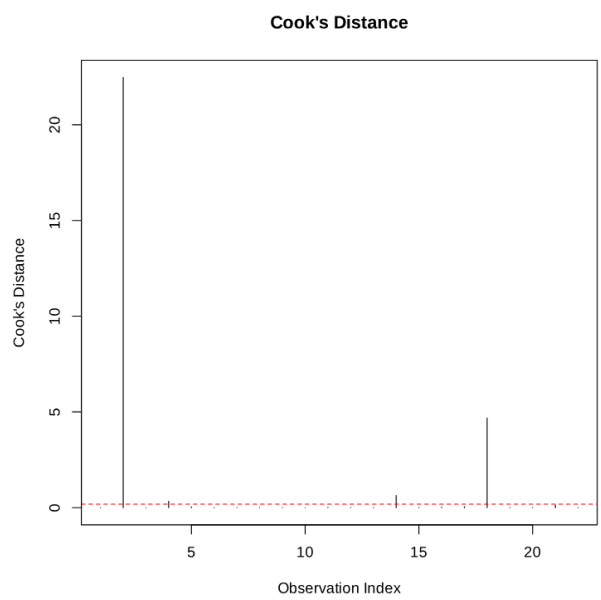

FIGURE 4.13—COOK'S DISTANCE OF PRIVATE-PD MODEL

Notes: Estimates show Index 2 & 18 as outliers.

A data.frame: 2 × 5

	Neighborhood	AI	PD	Private	Public_Art_Count
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
2	downtown	63251	62030	240	249
18	strathcona	31534	12585	177	50

FIGURE 4.14—THE OUTLIERS OF PRIVATE-PD MODEL

```
Call:
lm(formula = Public_Art_Count ~ Private * PD + I(PD^2), data = final_data[-c(2,
    18), ])

Residuals:
    Min      1Q  Median      3Q     Max
-19.154  -5.230  -1.426   5.029  20.506

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.912e+01  1.272e+01   1.503    0.154
Private      2.765e-01  7.258e-01   0.381    0.709
PD          -1.303e-03  1.052e-03  -1.239    0.234
I(PD^2)      2.686e-08  1.839e-08   1.460    0.165
Private:PD  -1.371e-06  2.221e-05  -0.062    0.952

Residual standard error: 11.74 on 15 degrees of freedom
Multiple R-squared:  0.5425,    Adjusted R-squared:  0.4205
F-statistic: 4.447 on 4 and 15 DF,  p-value: 0.01439
```

FIGURE 4.13—PRIVATE-PD MODEL

Notes: Excluding Downtown and Strathcona.

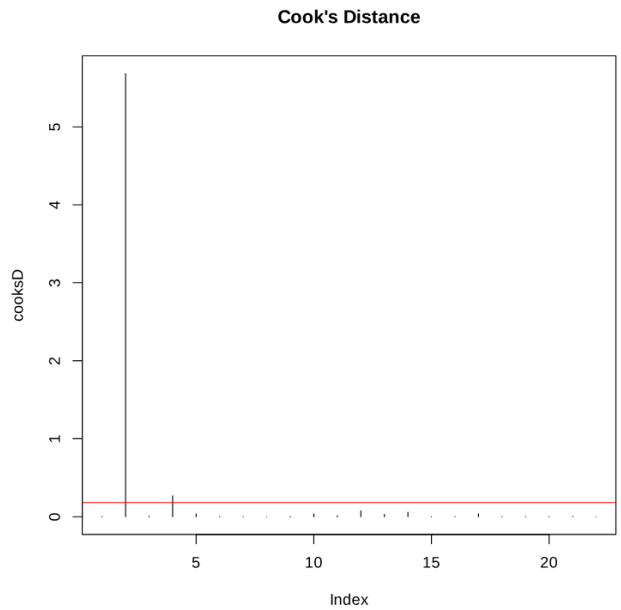

FIGURE 4.16—COOK'S DISTANCE OF REVISED MODEL

Notes: Estimates show Index 2 & 4 as outliers.

A data.frame: 2 × 5

	Neighborhood	AI	PD	Private	Public_Art_Count
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
2	downtown	63251	62030	240	249
4	fairview	61627	33620	137	21

FIGURE 4.17—THE OUTLIERS OF THE REVISED MODEL

```
Call:
lm(formula = Public_Art_Count ~ Private + AI + PD, data = final_data[-c(2,
    4), ])

Residuals:
     Min       1Q   Median       3Q      Max
-17.5664  -4.3584   0.3365   5.6949  20.7208

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.566e+01  1.108e+01  -2.316  0.03413 *
Private      2.955e-01  4.261e-02   6.935 3.36e-06 ***
AI           3.319e-04  1.226e-04   2.707  0.01555 *
PD           5.674e-04  1.886e-04   3.009  0.00833 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 9.263 on 16 degrees of freedom
Multiple R-squared:  0.7588,    Adjusted R-squared:  0.7136
F-statistic: 16.78 on 3 and 16 DF,  p-value: 3.391e-05
```

FIGURE 4.18—REVISED MODEL

Notes: Excluding Downtown and Strathcona.

A data.frame: 2 × 5

	GVIF	Df	GVIF^(1/(2*Df))	Interacts With	Other Predictors
	<dbl>	<dbl>	<dbl>	<chr>	<chr>
Private	1	4	1	PD	--
PD	1	4	1	I(PD^2), Private	--

FIGURE 4.19—PREDICTOR TYPE VIF FOR PRIVATE-PD MODEL

```
        Durbin-Watson test

data:  revised_model
DW = 1.9842, p-value = 0.4967
alternative hypothesis: true autocorrelation is greater than 0
```

FIGURE 4.20—DW TEST FOR REVISED MODEL

```
Durbin-Watson Test:
        Durbin-Watson test

data:  private_pd_model
DW = 1.9601, p-value = 0.4609
alternative hypothesis: true autocorrelation is greater than 0
```

FIGURE 4.21—DW TEST FOR PRIVATE-PD MODEL

```
Bootstrap Standard Errors:

ORDINARY NONPARAMETRIC BOOTSTRAP


Call:
boot(data = final_data, statistic = boot_model, R = 1000)


Bootstrap Statistics :
         original        bias     std. error
t1*  4.111208e+01 -1.133939e+01 1.781807e+01
t2* -2.149980e-02  4.556122e-02 4.500571e-01
t3* -3.185176e-03  9.775271e-04 1.565154e-03
t4*  5.617559e-08 -1.480036e-08 2.827173e-08
t5*  1.245364e-05 -4.205883e-06 1.542599e-05
```

FIGURE 4.22—BOOTSTRAP ESTIMATES & STANDARD ERRORS OF PRIVATE-PD MODEL

Notes: t1*- intercept, t2*- Private, t3*- PD, t4*- I(PD^2), t5*- Private:PD.

	Dependent variable:		
	Public Art Installations		
	(1)	(2)	(3)
Private	0.567***	0.492***	-0.021
	(0.105)	(0.088)	(0.107)
AI		0.001**	
		(0.0003)	
PD		0.002***	-0.003***
		(0.0005)	(0.001)
I(PD2)			0.00000***
			(0.00000)
Private:PD			0.00001***
			(0.00000)
Constant	2.323	-83.006***	41.112***
	(8.603)	(25.590)	(12.995)
Observations	22	22	22
R2	0.591	0.766	0.939
Adjusted R2	0.571	0.727	0.925
Residual Std. Error	34.261 (df = 20)	27.330 (df = 18)	14.318 (df = 17)
F Statistic	28.943*** (df = 1; 20)	19.638*** (df = 3; 18)	65.812*** (df = 4; 17)
Note:			*p<0.1; **p<0.05; ***p<0.01

FIGURE 4.23—TABLE OF RESULTS

Notes: Private-PD model (3) should refer to Figure 5 for detailed betas.

VI. Discussion

Cities are cultural landscapes where public art shapes communities and fosters shared identity. However, the factors influencing the distribution of public art are not well understood. While cultural equity is a common goal for policymakers, the role of private property is often overlooked. For example, why does a mural appear in one community but not another? This study explores the relationship between private cultural venues, population density, and public art installations.

A. Model 1

Simple Regression Interpretation.—Model 1 uses linear regression to examine how private cultural venue ownership affects public art density (1). The results reveal two primary observations.

A significant negative relationship exists between average household income and the number of public art installations (β = -0.001059; Figure 3.1). For every dollar increase in average household income, the number of public art installations decreases by 0.001059. This finding suggests that higher-income neighborhoods tend to have fewer public art installations, possibly due to a preference for curated or private artistic endeavors over public expressions of art. The t-value of −2.551 and p-value of 0.01903 confirm statistical significance at the 5% level ($p<0.05$).

The results for our dummy variable, which is the primary focus of our study, indicate an impact of −16.02 when $D_i=1$ (Figure 3.1). This suggests that neighbourhoods with a higher density of privately owned cultural spaces tend to have, on average, 16.02 fewer public art installations compared to neighbourhoods with government-owned spaces.

However, this variable does not strongly predict the number of public art installations. The t-value of −1.383 and p-value of 0.18188 ($p>0.05$) (Figure 3.1) do not provide sufficient evidence to reject the null hypothesis. This implies that the observed difference may not be statistically significant and could be due to random variation rather than a meaningful relationship. Given the inconclusive results, the model was modified to account for interactions between variables.

Complex Regression Interpretation.—This analysis examines how privately owned cultural spaces (D_i=1) affect the model variables, comparing them to government-owned spaces (D_i=0) (2). By including the ownership dummy, we assess how ownership type impacts public art installations and interacts with factors like density and income (Figure 3.2). We observe a significant improvement compared to the previous linear model. The F-statistic of 10.29 and its corresponding p-value of 6.831e-5 confirm a significant relationship being interpreted inside the model.

In explaining the number of public art installations, the model reveals several key insights regarding the relationship between density, average household income, and ownership type. Density ($p < 0.001$) has a strong positive relationship with the number of public art installations, as indicated by the coefficient of 2.141e+05 and a highly significant t-value of 6.204, confirming its robustness as a predictor (Figure 3.2). The density:average household income interaction ($p < 0.001$) shows a significant negative relationship, meaning the combined effect reduces the number of public art installations. This highlights the moderating role of income on the density-art relationship. Additionally, the density:average household income:ownership dummy interaction ($p = 0.00158$) suggests that the number of public art installations increases when all three variables interact. This three-way interaction outlines the complexity of the interaction between density, income, and ownership.

Overall, the interaction terms improve the model's explanatory power, highlighting the role of density as the strongest predictor, with income and ownership significantly moderating its effect on public art. The negative two-way interactions suggest that specific combinations of these predictors reduce public art installations. In contrast, the three-way interaction introduces a nuanced dynamic where their combined influence increases installations.

B. Model 2

The results from this estimation are both interesting and enlightening. Private ownership is a statistically significant predictor of the density of public art, net of the included socio-economic controls—Average Total Income (AI) and Population Density (PD). Adding the interaction terms and quadratic controls highlight the complicated relationship between PD and Private cultural venues, suggesting a dynamic that differs strongly between high and low density neighbourhoods.

The baseline regression (3), highlights the positive association between the presence of privately operated cultural venues and public art installations.To be more precise, the beta coefficient (β_1) of 0.5668*** (p-value <0.001) suggests that for each additional privately owned cultural venue in a neighborhood, there are approximately ~0.57 more public art installations (Figure 4.1). The relationship is strengthened by using robust standard errors and heteroskedasticity tests (Breusch-Pagan p-value = 0.000133) suggest the need for caution in interpretation (Figure 4.2). Building on this, the Revised Model adds controls for AI and PD, which notably enhance the explanatory power of the model ($Adjusted\ R^2$=0.727) (4)(Figure 4.3). In this connection, the coefficient for private ownership is reduced marginally to 0.4924*** (p-value <0.001), which remains strongly significant. Importantly, AI and PD also come out as strong predictors; the coefficient of PD is 0.001543 (p-value < 0.01), indicating that denser neighborhoods are likely to host more public art installations.

Robustness and Nonlinear Controls.—The addition of nonlinear elements, represented by AI^2 & PD^2 in the robustness framework, brings out interesting dynamics (5). The positive and statistically significant β_5 (8.747e-08**, p-value < 0.01) indicate a quadratic association, suggesting that population density has a low to negligible impact on public art at lower densities but becomes increasingly influential in areas with greater density (Figure 4.6). This result is essential for comprehending the interaction term (6)(Figure 4.7) and the result of the final Private-PD model ($Private{:}\,PD$), which presents a positive and highly significant coefficient (1.245e-05***, p-value< 0.001) (7)(Figure 4.8). This shows that in areas with higher population densities, the effect of private cultural institutions on the density of public art is strongly increased. Graphical representations confirm this parabolic relationship (Figure 4.9).

Sub-Sample and Sensitivity Analysis.—Heterogeneity becomes a noticeable theme in sub-sample analyses. In high-density neighborhoods—defined here as PD > 29,250 (Figure 2)—the coefficient for the interaction term doubles to 2.797e-05* (p-value < 0.05), further supporting this notion that private cultural venues become crucial in such cases (Figure 4.11). By contrast, in low-density neighborhoods, private ownership does matter (0.6658*, p-value < 0.05), but PD and its interaction term do not hold as explanatory (Figure 4.12). This divergence highlights how

population density mediates the relationship between private cultural venues and public art density.

The exclusion of major outliers, specifically Downtown and Strathcona, as identified by Cook's Distance, puts the fragility of these relationships in low-density settings into focus (Figure 4.13 & 4.14). More specifically, removing these most influential neighborhoods from the Private-PD Model causes every coefficient to lose all statistical significance, including those accounting for the interaction term— $Private{:}\,PD$—and for the quadratic population density variable—PD^2. This would suggest that previous findings of relationships are more generally driven by regions with high population density.

An especially interesting findings comes from the interaction term: (β_1) its beta changes to reflect a negative relation between private cultural venues and public art installation, which is in direct contrast to the positive effect of Private while controlling for PD (Figure 4.15). This discrepancy is further reinforced through the outlier analysis of the revised model, where Private remains of statistical significance; yet, its effect size considerably drops, falling from 0.49236 to 0.2955 (Figures 4.3, 4.16-4.18). Now, what does this mean? Our results support the former hypothesis that population density affects public art installations in a nonlinear, parabolic way. At low density levels, the impact of Private cultural venues appears close to trivial—nearly zero; but as density increases in magnitude, this relationship turns out to be stronger in its drive toward the significant interaction effect. The analysis of outliers further supports this assertion, in that for areas where private cultural institutions are scarce, the association with public art facilities drops to a level of statistical insignificance. While some positive association between private venues and public art can still be detected at lower levels of private activity, the data indicates that any such relationship is, at best, tenuous and primarily driven by urban areas with greater population densities.

Evaluation and Validation.—Diagnostic tests confirm the reliability of these findings. Variance Inflation Factors (VIF < 1.5 across all models) are indicative of very little multicollinearity (Figure 4.4, 4.19). Whereas heteroskedasticity is still present, robust standard errors mitigate the issue of betas being uninterpretable (Figure 4.2, 4.5, 4.8.) Durbin-Watson statistics (~2) confirm no evidence of autocorrelation, further validating the model (Figure 4.20, 4.21). Bootstrap methods, such as the "leave-one-out" analysis we ran give stronger confidence in the stability of

the coefficients. The interaction term is significant on every iteration, indicating that it is robust. Most important, the bias-corrected bootstrap estimates are quite close to the original point estimates, making the results more believable (Figure 4.22).

C. Comparison of Results

The results of both models reveal a complex relationship between public art installations, private cultural venues, population density, and other socioeconomic factors. This suggests that these variables interact dynamically rather than independently. Model 1 highlights that while average household income shows a significant inverse relationship with public art installations, the ownership dummy variable (private vs. government ownership) only weakly influences outcomes on its own. However, the inclusion of interaction terms in the complex regression underscores the critical interplay between population density, income, and ownership type, revealing that densely populated areas with higher-income neighborhoods experience a moderating effect on public art installations. Model 2 builds upon these findings, demonstrating a more complex understanding that private cultural venues have a positive and statistically significant effect on public art density. Particularly, there is a significance in high-density areas, where their influence intensifies as population density increases. This nonlinear, quadratic relationship, suggests that private cultural institutions become drivers of public art installations in urban contexts with higher population density, while their impact diminishes in less densely populated areas. Together, the models suggest that public art installations are not merely a function of individual variables but rather the result of mutual interactions between private ownership, density, and socioeconomic conditions.

D. Broader Implications

These research findings have salient implications for policy-practitioners and planners. In high-density cities, the findings hint that the promotion of private culture institutions seems to be very effective in increasing public artworks. For low-density neighborhoods however, a different approach—possibly emphasizing public or non-profit support—may be necessary. This brings us to the complicated function of population density as compelling a need for an adequate tailoring of policies at the level of specific features of neighborhoods. This analysis clarifies the complicated relationships between private cultural venues, population density, and public art,

contributing to a deeper understanding of urban cultural ecosystems. This study establishes a foundation for cultivating equitable and vibrant public art environments by revealing the conditions that allow private cultural venues to thrive.

VII. Conclusion

This paper set out to examine the relationship between the ownership of cultural venues and density of public art installations with a goal of understanding how these factors interact to shape urban cultural landscapes. The findings of this study underscore the significant role that private cultural venues play in shaping the distribution of public art installations, particularly in high-density neighborhoods. The final models revealed private cultural venues increase public art density, with a coefficient of 0.4924 ($p<0.001$). This was contrary to our initial expectation that public art is primarily driven by government initiatives.

Robustness checks and diagnostics tests reaffirmed these findings in high-density settings. Notably the relationship becomes even more pronounced in densely populated areas (nearly doubling the ownership effect). This highlights the importance of private institutions in fostering cultural vibrancy within urban spaces. These insights align with and expand upon the broader understanding of how cultural infrastructure contributes to community identity and social equity, offering valuable implications for policymakers seeking to balance public and private investments in urban cultural landscapes. Overall, our findings highlight the importance of tailoring urban cultural policies to account for these dynamics, particularly in encouraging formation of private cultural venues in high-income and high-density neighborhoods, and public art formulation in low-income and low-density neighborhoods for urban cultural development.

VIII. References

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IX. Attributions

(i) Fernando: Coded Model 1 regression model surrounding density and income. Wrote Discussion of Model 1.

(ii) Matthew: Coded Simple Model, Revised Model, Private-PD Model, Designed the 3 Models, Discussion section write up, Specification Check code and write up, Robustness Check code and write up, Model Section Write Up, Summary Statistic Section Write Up, Table of Results Section Write up.

(iii) Alastair: Coded Cook's Distance Influence Estimate Graph and residual Histogram. Wrote introduction, comparison of models and portions of conclusion.

(iv) Mollie: Helped with formatting as well as general grammatical edits throughout the paper. Wrote conclusion and census data description. Coded White's test for Heteroskedasticity and Effect of PD on Public Art Count Graph.

(v) Natasha: Coded "Residual v.s Fitted Values Graph" and "Private Cultural Venues vs Public Art Count" Graph. Wrote cultural venues and public art installation data descriptions. Completed general edits and formatted document into AEA-AEJ.